

THE OCEANS: CURRENTS, TIDES, SALINITY / INDIAN OCEAN AND IOD

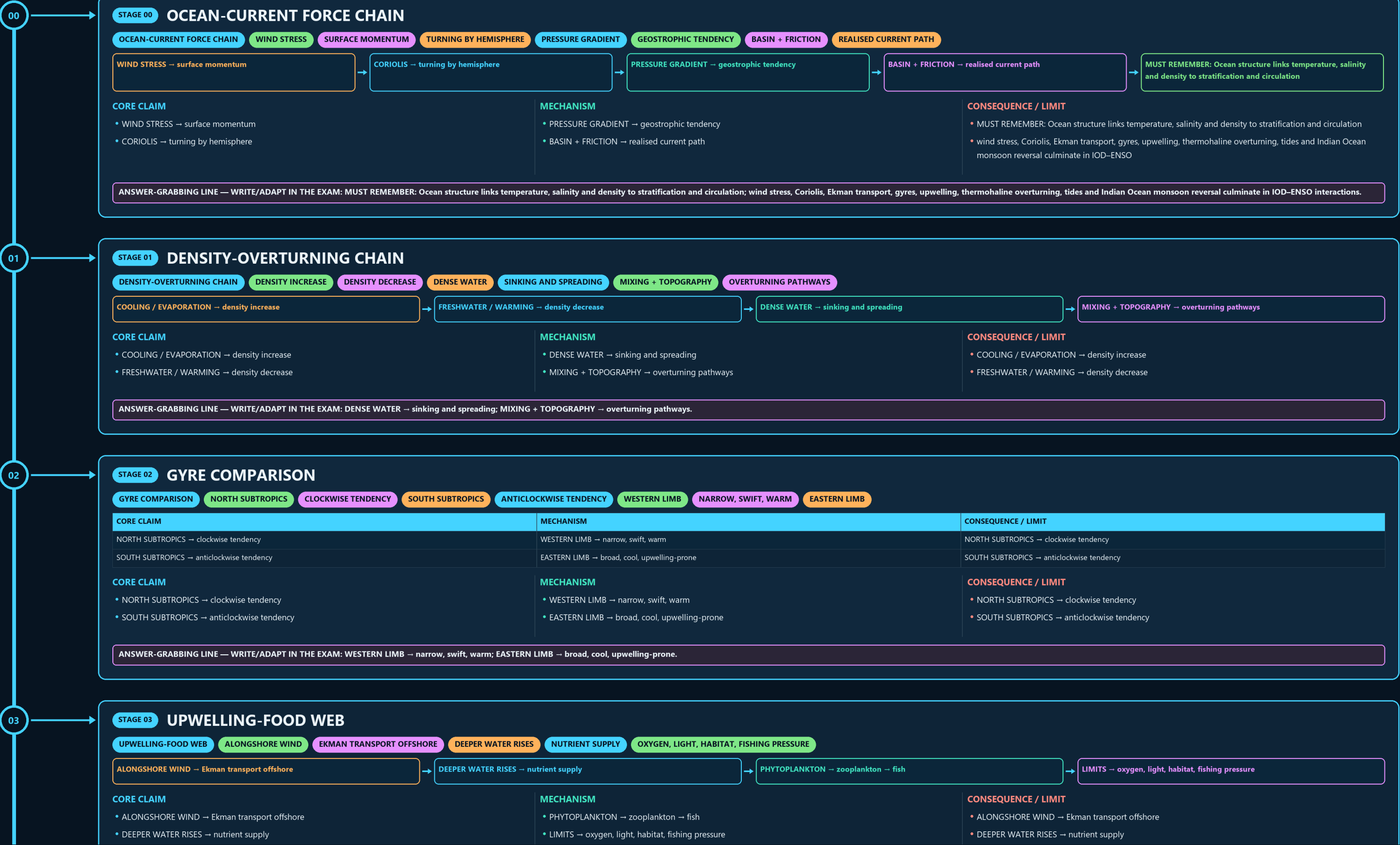
OCEAN-CURRENT FORCE CHAIN → DENSITY-OVERTURNING CHAIN → GYRE COMPARISON → UPWELLING-FOOD WEB → TIDE GEOMETRY → LOCAL TIDAL RANGE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WESTERN LIMB → narrow, swift, warm; EASTERN LIMB → broad, cool, upwelling-prone.

03

STAGE 03 UPWELLING-FOOD WEB

UPWELLING-FOOD WEB ALONGSHORE WIND EKMAN TRANSPORT OFFSHORE DEEPER WATER RISES NUTRIENT SUPPLY OXYGEN, LIGHT, HABITAT, FISHING PRESSURE

ALONGSHORE WIND → Ekman transport offshore

DEEPER WATER RISES → nutrient supply

PHYTOPLANKTON → zooplankton → fish

LIMITS → oxygen, light, habitat, fishing pressure

CORE CLAIM

- ALONGSHORE WIND → Ekman transport offshore
- DEEPER WATER RISES → nutrient supply

MECHANISM

- PHYTOPLANKTON → zooplankton → fish
- LIMITS → oxygen, light, habitat, fishing pressure

CONSEQUENCE / LIMIT

- ALONGSHORE WIND → Ekman transport offshore
- DEEPER WATER RISES → nutrient supply

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PHYTOPLANKTON → zooplankton → fish; LIMITS → oxygen, light, habitat, fishing pressure.

04

STAGE 04 TIDE GEOMETRY

TIDE GEOMETRY FULL MOON ALIGNED FORCES SPRING TIDE LARGER RANGE QUARTER MOON PARTIAL OFFSET NEAP TIDE

CORE CLAIM

NEW / FULL MOON → aligned forces
SPRING TIDE → larger range

MECHANISM

QUARTER MOON → partial offset
NEAP TIDE → smaller range

CONSEQUENCE / LIMIT

NEW / FULL MOON → aligned forces
SPRING TIDE → larger range

CORE CLAIM

- NEW / FULL MOON → aligned forces
- SPRING TIDE → larger range

MECHANISM

- QUARTER MOON → partial offset
- NEAP TIDE → smaller range

CONSEQUENCE / LIMIT

- NEW / FULL MOON → aligned forces
- SPRING TIDE → larger range

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUARTER MOON → partial offset; NEAP TIDE → smaller range.

05

STAGE 05 LOCAL TIDAL RANGE

LOCAL TIDAL RANGE ASTRONOMICAL FORCING INCOMING TIDE SHELF DEPTH + BAY SHAPE RESONANCE + FRICTION TIMING AND HEIGHT SAME FORCING, DIFFERENT LOCAL RANGE

ASTRONOMICAL FORCING → incoming tide

SHELF DEPTH + BAY SHAPE → amplification

RESONANCE + FRICTION → timing and height

RULE → same forcing, different local range

CORE CLAIM

- ASTRONOMICAL FORCING → incoming tide
- SHELF DEPTH + BAY SHAPE → amplification

MECHANISM

- RESONANCE + FRICTION → timing and height
- RULE → same forcing, different local range

CONSEQUENCE / LIMIT

- ASTRONOMICAL FORCING → incoming tide
- SHELF DEPTH + BAY SHAPE → amplification

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESONANCE + FRICTION → timing and height; RULE → same forcing, different local range.

06

STAGE 06 SALINITY BUDGET

SALINITY BUDGET EVAPORATION + BRINE SALINITY RISES RAIN + RIVER + MELT SALINITY FALLS ADVECTION + MIXING REDISTRIBUTE SALT TEMP + SALINITY

EVAPORATION + BRINE → salinity rises

RAIN + RIVER + MELT → salinity falls

ADVECTION + MIXING → redistribute salt

TEMP + SALINITY → density and stratification

CORE CLAIM

- EVAPORATION + BRINE → salinity rises
- RAIN + RIVER + MELT → salinity falls

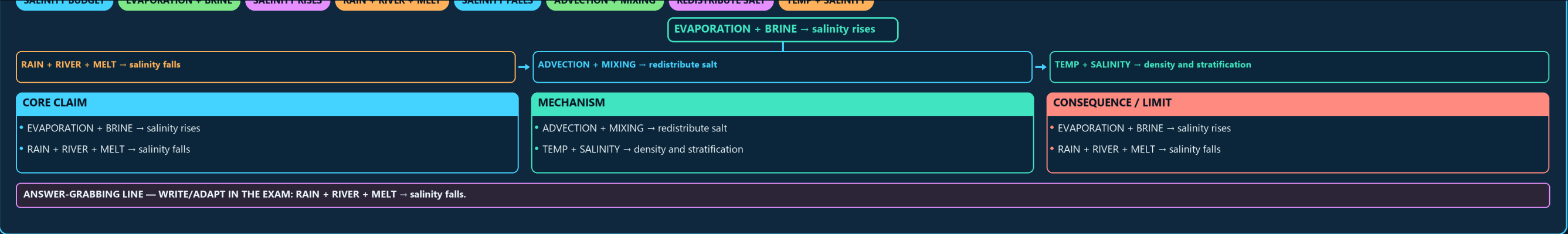
MECHANISM

- ADVECTION + MIXING → redistribute salt
- TEMP + SALINITY → density and stratification

CONSEQUENCE / LIMIT

- EVAPORATION + BRINE → salinity rises
- RAIN + RIVER + MELT → salinity falls

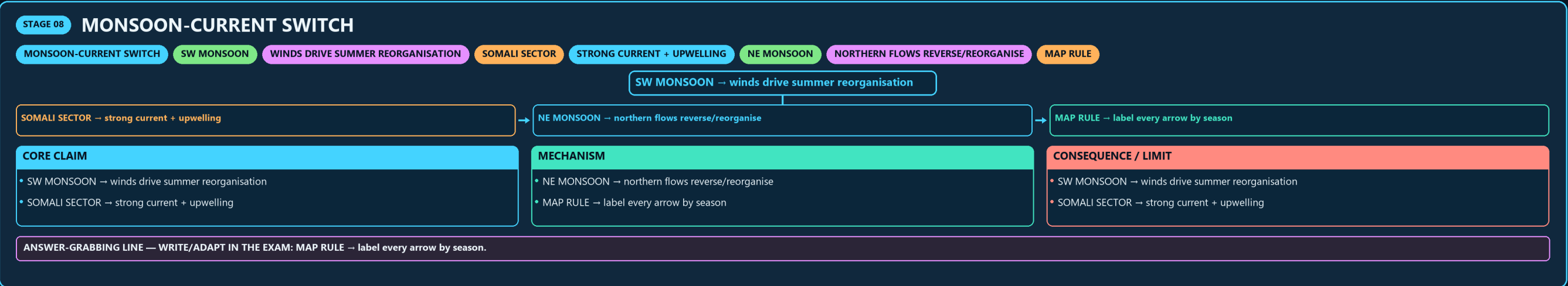
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RAIN + RIVER + MELT → salinity falls.



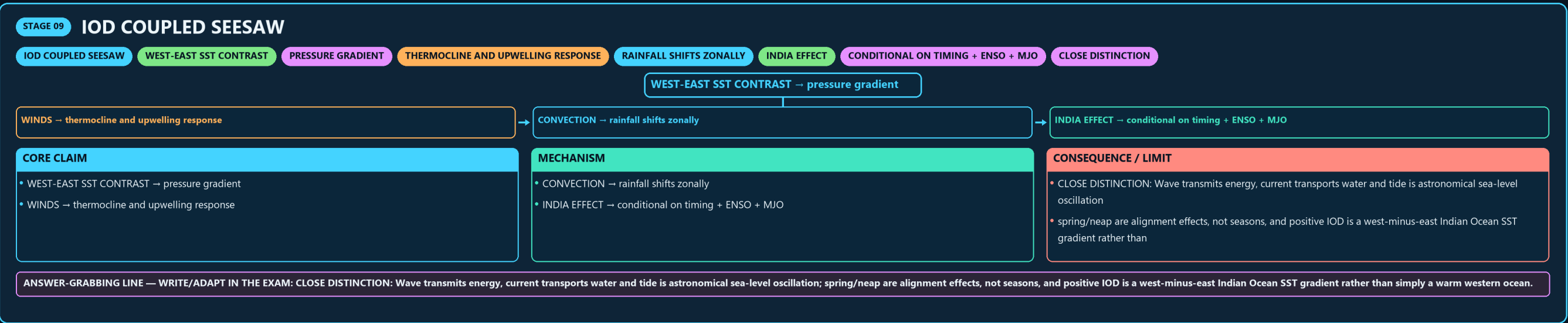
07

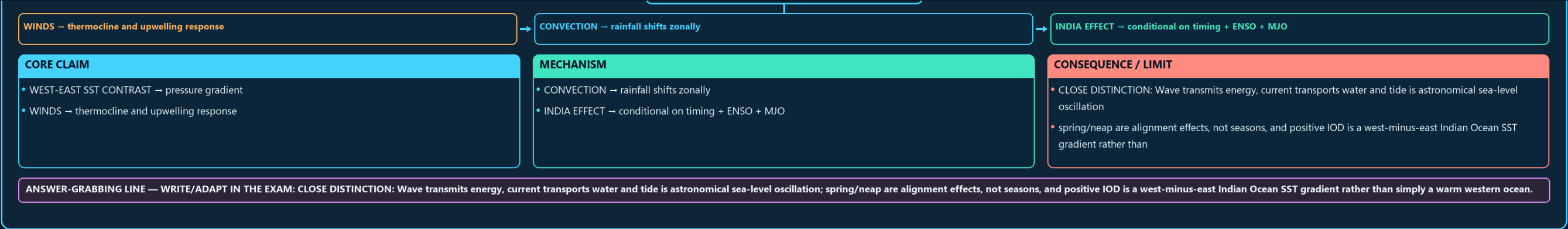


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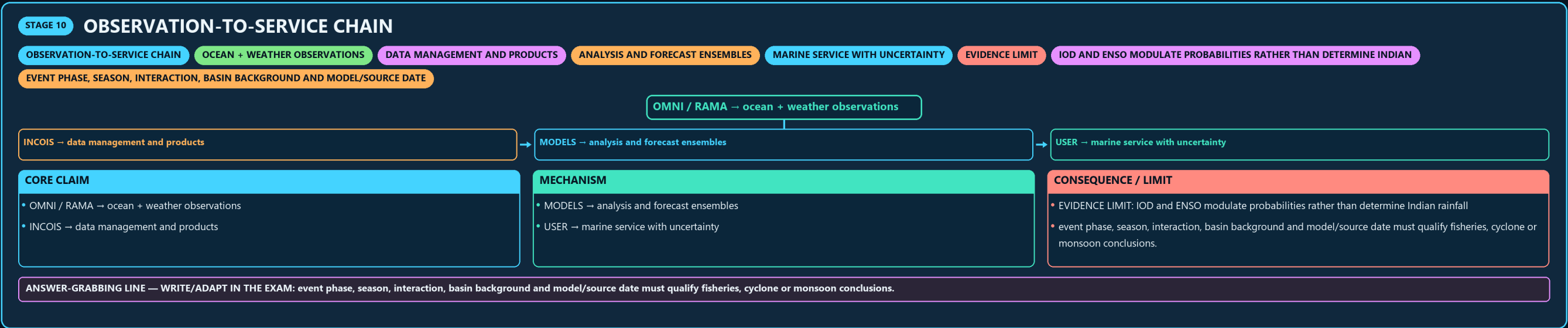


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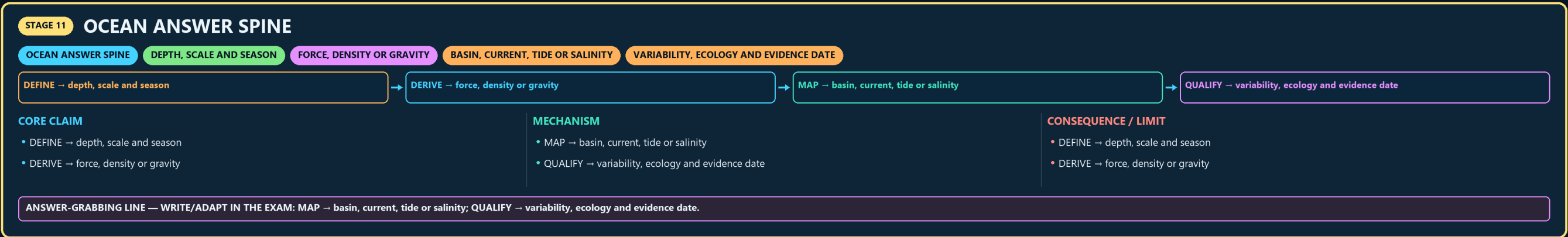




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11



E

